MEMS-Based Flexible Electrode Array and Impedance-Adaptive Closed-Loop Wireless System for Selective Deep-Brain Stimulation with Non-Invasive Temporal Interference

[1,2]Ji-Hyoung Cha*,[1,2]Shinwoo Lee*, [1]Keuntae Kim, [3]Hyogeun Shin, [4]Namsun Chou, [5]Jiwan Woo & [1,2]Il-Joo Cho*
[1]Department of Convergence Medicine, College of Medicine, Korea University, Seoul, Republic of Korea
[2]Department of Anatomy, Brain Korea 21 Plus Program for Biomedical Science, College of Medicine, Korea University, Seoul, Republic of Korea [3]School of Electronics Engineering, College of IT Engineering, Kyungpook National University, Daegu, Republic of Korea [4]Emotion, Cognition, & Behavior Research Group, Korea Brain Research, Daegu, Republic of Korea [5]Research Animal Resources Center, Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea
E-mail: ijcho@korea.ac.kr

비침습적 시간 간섭 기반의 선택적 뇌심부 자극을 위한 MEMS 기반 유연 전극 어레이와 임피던스 적응형 폐루프 무선 시스템

[1,2]차지형*, [1,2]이신우*, [1]김근태, [3]신효근, [4]추남선, [5]우지완 & [1,2]조일주*
[1]고려대학교 의과대학 융합의학교실 [2]고려대학교 의과대학 해부학교실 BK21 플러스 프로그램
[3]경북대학교 IT 대학 전자공학부 [4]한국뇌연구원 정서·인지질환 연구그룹 [5]한국과학기술연구원 연구동물자원센터
E-mail: ijcho@korea.ac.kr

Abstract

Non-invasive temporal interference stimulation delivers two high-frequency carriers with a small frequency offset to generate a low-frequency envelope, enabling selective deep-brain activation while minimizing superficial stimulation. However, conventional wired systems impose behavioral constraints and suffer from unstable electrode impedance, as motion-induced impedance variations compromise stimulation stability. We present a small-form-factor, bidirectional wireless system featuring impedance-adaptive closed-loop feedback and MEMS-based flexible electrode array. The platform actively compensates for dynamic impedance variations, ensuring stable constant-current delivery. This miniaturized system enables selective region targeting among multiple brain regions in freely moving mice, preserving the interference field for precise neuromodulation. This work establishes a scalable foundation for chronic, non-invasive deep-brain stimulation without compromising natural behavior.

Keywords: ***Temporal interference stimulation, bidirectional wireless system, small-form-factor, impedance-adaptive closed-loop feedback, selective region targeting***

1. Introduction

Temporal interference stimulation (TIS) is a promising non-invasive technique that utilizes two high-frequency carriers to create a low-frequency interference envelope, enabling the selective activation of neurons within deep-brain regions. However, traditional TIS configurations rely on bulky, wired benchtop architectures [1], which restricts stimulation to a single target region and necessitates manual hardware reconfiguration for each experimental paradigm. In addition, dynamic impedance variations caused by micromotion and contact instability hinder constant-current stimulation, ultimately compromising both the reliability of stimulation and animal safety [2].

We address these limitations by introducing a miniaturized wireless impedance-adaptive closed-loop TIS system. As shown in Fig. 1, this architecture integrates real-time current sensing with dynamic compliance voltage control, ensuring stable stimulation in freely moving animals despite impedance fluctuations. Furthermore, a microelectromechanical system (MEMS)-based flexible microelectrode array (MEA) supports reconfigurable and selective region targeting. To our knowledge, this work represents the first miniaturized wireless TIS platform to achieve in vivo constant-current stimulation with real-time impedance-adaptive feedback. By integrating closed-loop regulation with selective region targeting, this system bridges the gap between the theoretical potential of TIS and the practical requirements for scalable and reliable neuromodulation.

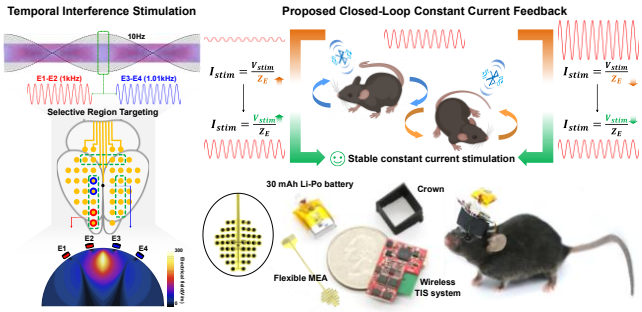

Fig. 1. Overview of MEMS-based wireless TIS system. Conceptual art of impedance-adaptive closed-loop constant-current feedback and photograph of the head-mounted system with flexible MEA.

2. Design of the wireless TIS system and MEMS-based flexible MEA

Our compact head-mounted system integrates TIS waveform generation, constant-current regulation, and wireless control. Sinusoidal signals are synthesized by the microcontroller unit (MCU) using two pulse-width modulation (PWM) channels and amplified to approximately 30 V_{pp}, providing sufficient headroom to drive currents from hundreds of µA to several mA. This strategy generates two precise carrier frequencies at 1 kHz and 1.01 kHz, producing a target envelope modulation frequency of 10 Hz in the brain. For closed-loop feedback, the stimulation current is monitored via a shunt resistor and digitized through a 12-bit analog-to-digital converter (ADC). The MCU then dynamically adjusts the PWM duty cycle to regulate the compliance voltage, ensuring stable current delivery despite impedance variations.

Bluetooth low energy (BLE)-based custom graphical user interface (GUI) supports real-time visualization and simultaneous multi-subject control. Selective region targeting, remotely controlled by GUI, allows rapid repositioning of stimulation sites without hardware reconfiguration. Weighing only 2.37 g (including a 30-mAh battery) with a 310 mm^2 footprint, the final device minimizes mechanical load on mice, preserving natural behaviors during chronic experiments.

The flexible MEA was fabricated using a MEMS-based polyimide (PI) multilayer process. Ti/Au interconnects were patterned via UV lithography between two 5-µm PI layers, with selective etching exposing the electrodes. This expanded multi-electrode array configuration was specifically engineered to enable simultaneous, high-precision selective region targeting across the mouse brain. The final array was released from the sacrificial layer, ensuring conformal placement on the skull surface.

3. In vivo validations of system under dynamic conditions in freely moving mice

To verify the performance of our wireless TIS system, we placed mice in an open-field box and applied the secondary motor cortex (M2) stimulation. The stimulation produced statistically significant rotational biases, consistent with previous studies [3-4], with clockwise and counterclockwise rotations for left and right M2, respectively. These behavioral modulations were validated via simultaneous target switching across multiple subjects (Fig. 2a-c). The rapid reversal of responses upon wireless commands confirms the minimal latency of our bidirectional architecture, ensuring synchronized real-time control over targeted regions in parallel.

We systematically evaluated the robustness of the system against inter-subject impedance variations and longitudinal stability (Fig. 2d). Despite a wide range of natural impedance fluctuations recorded from electrode pairs in each hemisphere (left or right), the system consistently elicited direction-specific behavioral modulations with high reproducibility. These results demonstrate that the closed-loop constant-current feedback actively compensates for dynamic impedance variations in real-time, ensuring stable TIS operation and long-term targeting fidelity.

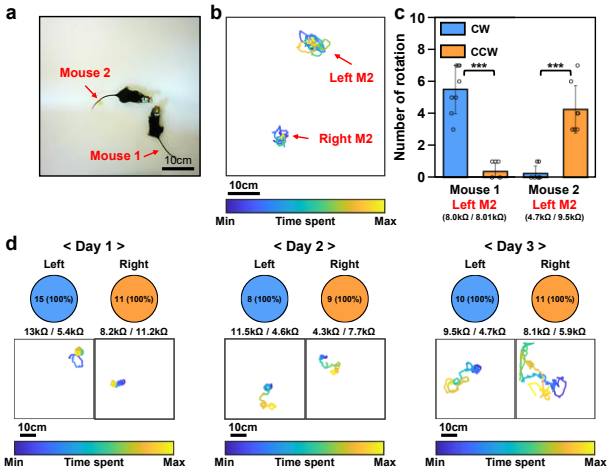

Fig. 2. System robustness under simultaneous multi-subject control. (a) Setup in the open-field box, (b) locomotive trajectories during TIS, (c) quantitative comparison of rotational behavior and measured impedance values, and (d) longitudinal stability of the wireless TIS system over three consecutive days. Unilateral stimulation sessions targeting the left and right M2 were performed each day under impedance monitoring; impedance values correspond to electrode pairs in each hemisphere.

4. Conclusion

We developed a miniaturized, bidirectional wireless TIS system with impedance-adaptive closed-loop control. Utilizing MEMS-based flexible MEAs and real-time sensing, the platform ensures stable constant-current delivery despite impedance fluctuations. Selective region targeting was validated by reproducible rotational behaviors during simultaneous multi-subject control. This scalable architecture enables chronic, non-invasive deep-brain stimulation without compromising natural behavior.

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